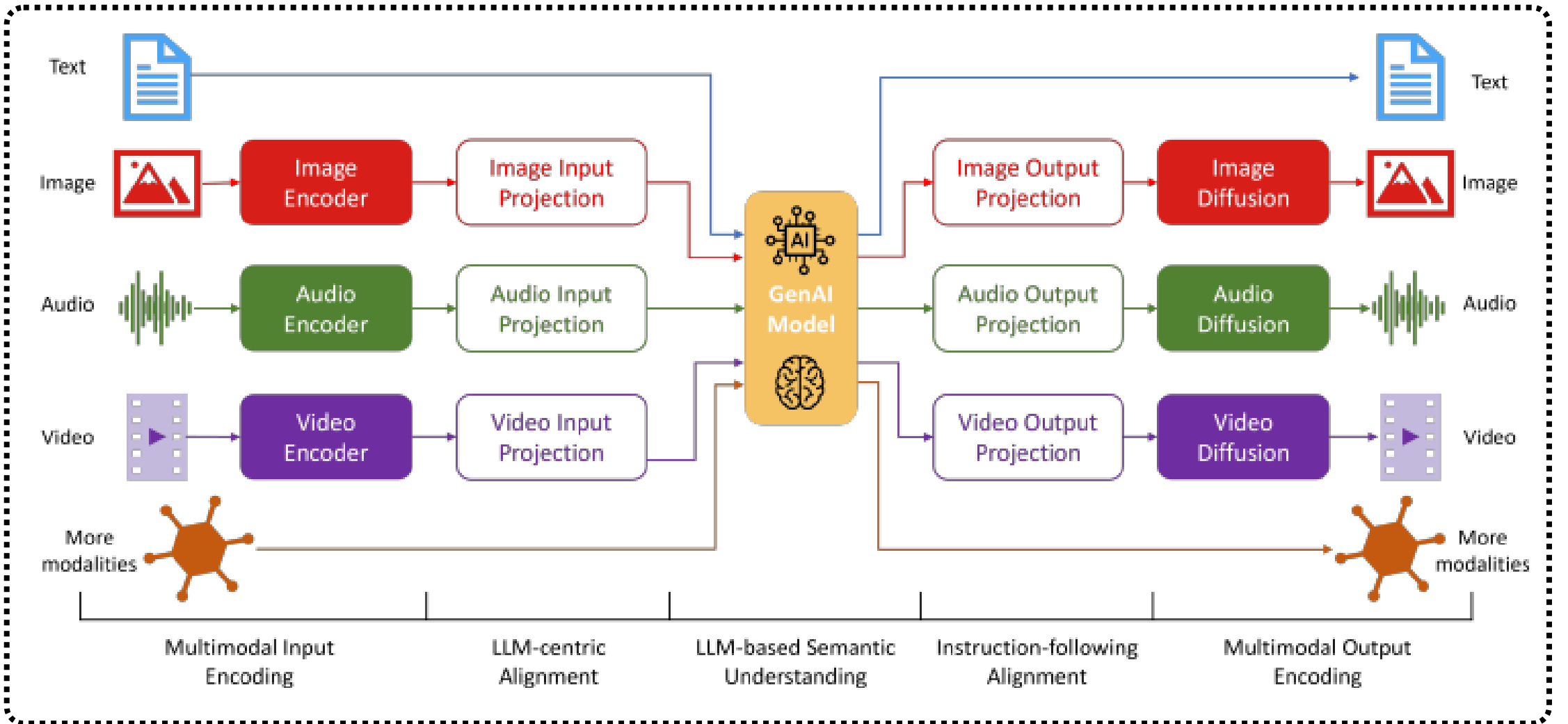


Multi-Modal GenAI for Energy Infrastructure Inspection Reports



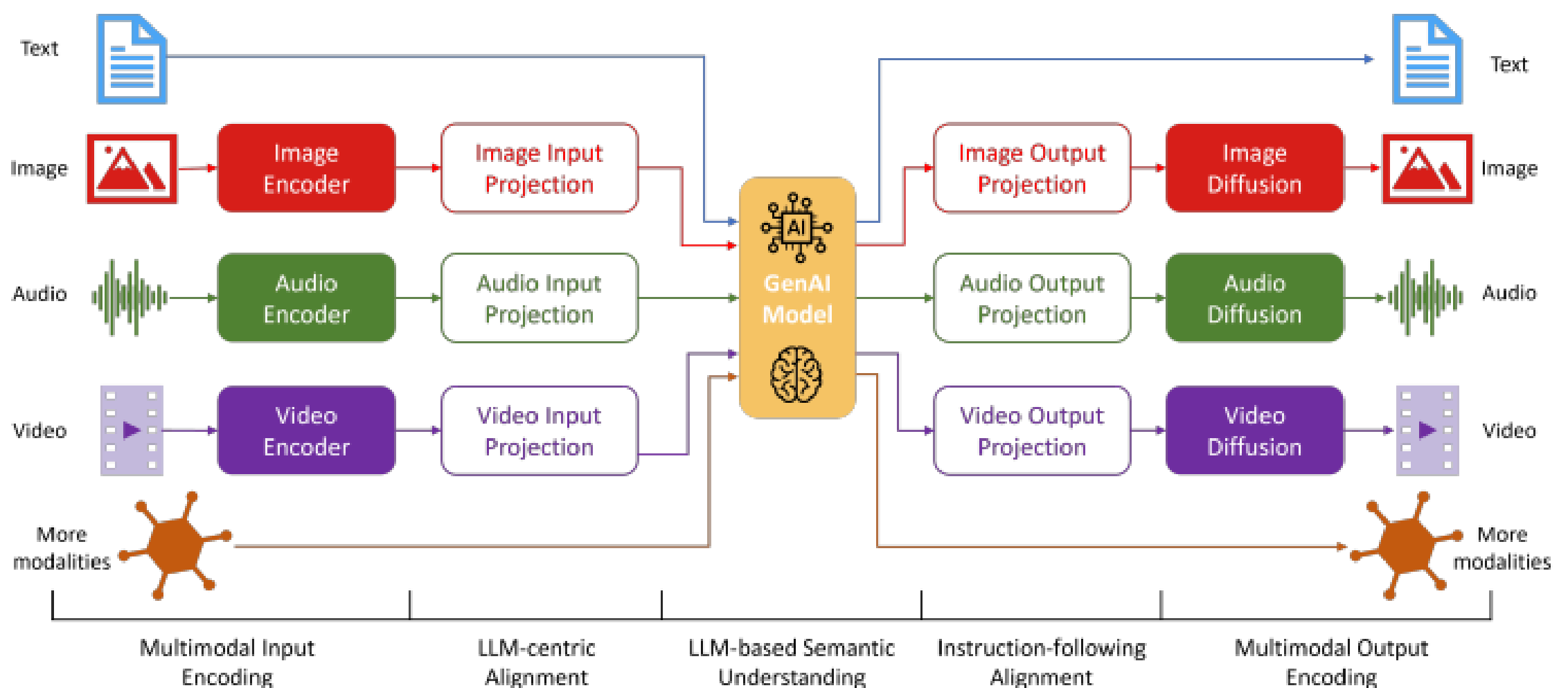
Task	Tools / Models
Image Captioning	BLIP, MiniGPT-4, Gemini Vision API, HuggingFace Vision-Language models
Report Generation	GPT-4, LLaMA 3, LangChain
Data Management (optional)	FAISS, Chroma for vector search
UI	Streamlit, Gradio
Data Format	Drone/CCTV images + inspection logs (CSV/JSON)

As the energy sector shifts towards renewable sources and connected systems, one thing becomes clear: **traditional inspection workflows can't keep up.**

Manual report generation from thousands of drone images and technician notes?

Not scalable.

Enter **Multi-Modal Generative AI**



This session dives into how we can use GenAI models that combine vision and language models understanding to automatically generate maintenance reports from raw inspection data think drone imagery of solar panels or text logs from field operators.

Hack Session

Multi-Modal GenAI for Energy Infrastructure Inspection Reports

Prakalp Somawanshi

Principal AI Engineer



What You'll Learn

This is a hands-on session **less theory, more doing**. You'll walk away knowing how to:

- **Understand Multi-Modal GenAI:** Learn how combining vision + language models changes the game for inspection automation.
- **Caption Inspection Images:** Use models like BLIP, MiniGPT-4, or Gemini Vision API to describe pipeline cracks, panel defects, or corrosion points in plain English.
- **Engineer Prompts that Work:** Feed image captions + technician notes into GPT-4 or LLaMA 3, guiding them with smart prompts to output clear, structured maintenance reports.
- **Build an Inspection Assistant:** Combine vision, language, and domain-specific cues to create an assistant that understands energy infrastructure, not just generic content.
- **Deploy a Functional App:** Use Streamlit or Gradio to create a working interface that lets users upload images and logs, then get a complete, AI-generated inspection summary.
- **Prototype with Open Tools:** Work with synthetic data, CSV logs, and open-source models no black boxes.

Technologies You'll Use

Task	Tools / Models
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